ST. THOMAS INSTITUTE FOR SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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SENTINEL AI — REAL-TIME FINANCIAL FRAUD INTELLIGENCE

Specialized Academic Report: Machine Learning Architecture & Hybrid Fraud Scoring

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Report Summary & Reading Guide:
This academic report provides an exact, accessible explanation of the real-time machine learning and hybrid risk scoring architecture implemented in the Sentinel AI fraud intelligence platform. All descriptions, formulas, feature sets, and thresholds are verified directly from active source files (src/models/hybrid_scorer.py, src/features/canonical_features.py, src/rules/fraud_rules.py, and config/config.py). The technical details are presented in simple, non-technical English to ensure clear conceptual understanding without sacrificing engineering rigor.

Document Type: B.Tech Project Report **Target Audience:** Non-Technical / Academic **Source of Truth:** Verified Codebase

HOW MY RANDOM FOREST PREDICTS WHETHER A TRANSACTION IS GENUINE OR FRAUDULENT

1. What Random Forest Means in Simple English

A **Decision Tree** is like a flowchart of simple yes-or-no questions. For example, it might ask: *"Was this card used in another country ten minutes ago?"* If yes, follow one path; if no, follow another. However, a single tree can make mistakes when transactions are complex. A **Random Forest** solves this by combining an entire committee (a "forest") of **100 independent decision trees**. Each tree looks at the transaction from slightly different angles, and they all vote on the final answer. By pooling their votes together, the forest eliminates individual errors and makes a reliable, unbiased prediction.

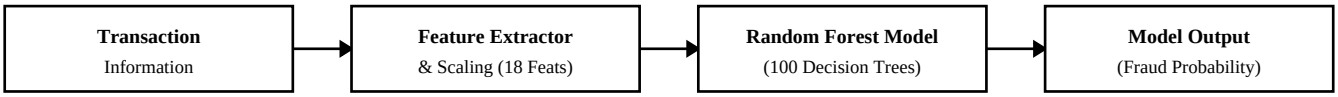
2. What Transaction Information the Model Receives

The model does not rely on obscure, uninterpretable codes. Instead, it receives **18 clear, observable features** grouped into six logical categories:

- **Transaction Size:** The transaction amount and its natural logarithm (to handle extreme purchases smoothly).
- **Time & Day:** Time of day (encoded smoothly as continuous cycles) and whether the transaction occurs on a weekend.
- **Spending Velocity:** How many transactions occurred on this card in the last 1 minute, 5 minutes, and 24 hours, plus total 24-hour spend.
- **Customer Spending Norms:** How much current spending deviates from the customer's average (spending ratio and statistical z-score).
- **Travel & Speed:** The physical distance (in kilometers) and travel speed (km/h) between this transaction and the previous purchase.
- **Familiarity & Merchant Risk:** Whether the merchant or city is new to the cardholder, plus industry risk weights for Merchant Category Codes (e.g., gambling: 0.95 vs. grocery: 0.10) and transaction channels (e.g., phone order: 0.70 vs. chip card: 0.10).

3. How the Actual Project Prepares Features & Evaluates the Transaction

When a transaction arrives, the CanonicalFeatureExtractor strictly reviews past events to calculate the 18 numbers. Next, a standard mathematical scaling tool (StandardScaler) adjusts all numbers so that large figures (like an $8,000 purchase) do not drown out small indicators (like a 5-minute transaction count). The 100 decision trees (each constrained to a maximum depth of 10 questions) then evaluate the scaled numbers simultaneously. Each tree casts a vote: **1 for Fraud** or **0 for Genuine**.



4. How Model Output Is Produced and Interpreted in Sentinel AI

Rather than forcing a rigid yes/no verdict, the Random Forest calculates the **exact fraction of trees that voted fraud**. For example, if 84 out of 100 trees vote fraud, the output is a **fraud probability of 0.84 (84%)**.

- **Standalone ML Benchmark:** During offline testing, transactions with probability ≥ 0.50 are classified as fraud.
- **Role in Live Sentinel AI:** In the actual real-time platform, the Random Forest does *not* make the final decision alone. Its output becomes the **ML Score [0.0 to 1.0]**, contributing a 50% weight to the comprehensive hybrid scoring pipeline.

5. Illustrative Example (Step-by-Step)

Illustrative Scenario (Hypothetical Cardholder Example):
A cardholder who usually spends $50 at local grocery stores in New York suddenly has a $2,400 online transaction at a gambling site (MCC 7995) initiated from London 12 minutes later.

1. **Feature Extraction:** Implied travel speed = 27,850 km/h; Amount ratio = 48.0x normal; Channel = Online; MCC risk = 0.95.
2. **Tree Evaluation:** Trees ask: *Is speed > 800 km/h?* (Yes). *Is amount > 10x average?* (Yes). *Is MCC high-risk?* (Yes).
3. **Model Output:** 98 out of 100 trees vote fraud → **ML Probability Score = 0.98 (98.0%)**.

HOW THE COMPLETE SCORING SYSTEM WORKS

Sentinel AI combines three independent layers into a unified defense system. Relying on machine learning alone can result in blind spots, while relying solely on rules lacks statistical adaptability. The complete scoring system synthesizes three specialized scores: **ML Scoring**, **Behavior Scoring**, and **Rule Scoring**.

1. The Three Scoring Components (Verified Implementation)

- **1. ML Scoring (Weight = 0.50 / 50%):** Produced by the production Random Forest classifier. It provides a statistical fraud probability [0.0 to 1.0] reflecting non-linear patterns, historical velocities, and high-risk merchant/channel associations across 18 canonical features.
- **2. Behavior Scoring (Weight = 0.30 / 30%):** Calculated by the BehaviorScorer, which compares the event against the customer's personal baseline. It evaluates: spending deviation (amount > 1.8x adds +0.18, > 3.0x adds +0.35); 5-minute velocity (> 3 txns adds +0.15, > 5 txns adds +0.30); geographic diversity (≥ 2 distinct cities in 1 hour adds +0.20); unfamiliar merchant (+0.10); unusual location (+0.12); unusual hours (+0.08); and 24-hour spending surge (> 2.5x normal adds +0.15). The components are summed and capped at 1.0.
- **3. Rule Scoring (Weight = 0.20 / 20%):** Evaluated by the FraudRuleEngine across seven explicit deterministic fraud rules: *Rule 1*: Burst (> 5 txns in 2 min, weight 0.20); *Rule 2*: Impossible travel (> 300 km in 30 min, weight 0.25); *Rule 3*: Spike (> 3x average amount, weight 0.15); *Rule 4*: New merchant with high amount (> 2x average, weight 0.15); *Rule 5*: Card testing (≥ 2 declines followed by high-value success, weight 0.10); *Rule 6*: Merchant card swarm (≥ 4 cards in 5 min, weight 0.05); *Rule 7*: Unusual transaction hour (weight 0.10). Triggered weights sum to a normalized score [0.0 to 1.0].

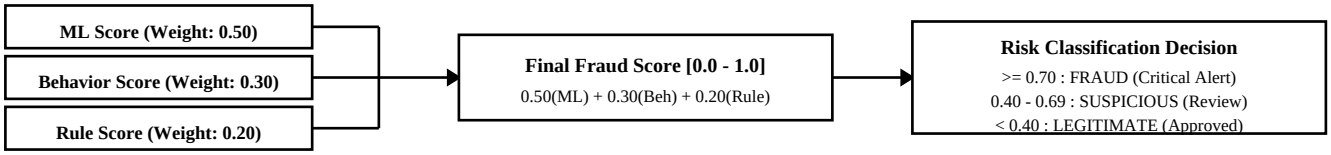
2. Mathematical Combination Formula & Decision Thresholds

In src/models/hybrid_scorer.py, the final fraud risk score is computed using the strict weighted formula:

Final Fraud Score = (0.50 × ML Score) + (0.30 × Behavior Score) + (0.20 × Rule Score)

All component scores are bounded between 0.0 and 1.0. The final result is rounded to 4 decimal places.

Score Range	Classification	Operational Meaning & System Action
0.70 to 1.00	**FRAUD**	Critical risk. Transaction is flagged/blocked. Real-time alert dispatched to security team.
0.40 to 0.69	**SUSPICIOUS**	Elevated risk. Triggers step-up authentication (OTP/biometrics) or queues for human review.
0.00 to 0.39	**LEGITIMATE**	Normal spending behavior. Transaction is automatically authorized with zero customer friction.



3. Post-Scoring Pipeline: Deduplication, Persistence & Explainability

Once the decision is rendered, Sentinel AI executes three critical operational steps:

- **Idempotency Guard:** The system verifies the transaction ID and idempotency key against a 30-minute memory cache. Duplicate network events return previous results without corrupting customer spending baselines.
- **Decoupled Database Persistence:** The full record (all 3 component scores, final score, triggered rules, and decision) is persisted to PostgreSQL/Supabase and memory buffers.
- **Human-Readable Explainability:** The system consolidates plain-English reasons (e.g., *"Impossible travel alert | Amount is 4.2x customer average"*), enabling compliance officers to audit decisions instantly.